

Virtual Data Science with Python Apprentice Intern

WEEK 5: FINAL PROJECT REPORT

Comprehensive Data Science Project Reporting and Strategic Recommendations

Project: Titanic Survival Analysis

Dataset: Titanic Dataset

Source: Kaggle

Technologies Used:

Python, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn

Machine Learning Model:

Logistic Regression

Statistical Test:

Chi-Square Test of Independence

Internship:

Virtual Data Science with Python Apprentice Intern

Submitted By:

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1. Executive Summary

This comprehensive project report presents the complete data science workflow carried out during the five-week Virtual Data Science with Python Apprentice Internship. The project used the Titanic dataset obtained from Kaggle and integrated the skills developed through data acquisition, data cleaning, exploratory data analysis, visualization, statistical analysis, hypothesis testing, and machine learning.

During Week 1, the dataset containing **891 passenger records and 12 columns** was acquired and examined using Python, Pandas, Matplotlib, and Seaborn. Data quality checks identified **177 missing values in Age, 2 missing values in Embarked, and 687 missing values in Cabin**. Age values were handled using the median, missing Embarked values using the mode, and the Cabin column was removed because of its large number of missing values. No duplicate records were found.

During Week 2, different visualizations were created to identify patterns in passenger survival, gender, passenger class, age, fare, and embarkation. These visualizations helped convert numerical data into understandable graphical insights.

During Week 3, statistical hypothesis testing was performed to investigate the association between passenger gender and survival. The Chi-Square Test of Independence produced a statistic of approximately **260.717** with **1 degree of freedom** and a p-value of approximately **1.197×10^{-58}** . Since the

p-value was below the significance level of 0.05, the null hypothesis was rejected.

During Week 4, a **Logistic Regression** model was developed using Pclass, Sex, Age, and Fare as features. After preprocessing, the working dataset contained **714 records**. The model used an 80/20 train-test split and achieved approximately **75.52% accuracy** with an **AUC of 0.81**.

The final analysis integrates these findings to demonstrate how data cleaning, visualization, statistical evidence, and predictive modelling can work together in an end-to-end data science project. Based on the findings, strategic recommendations, potential business/research impacts, limitations, and future improvement areas are discussed. The project demonstrates the importance of using multiple analytical approaches rather than relying on a single technique when deriving insights from data

2.Methodology

The project followed a structured end-to-end data science workflow in which the learning and outputs from Weeks 1 to 4 were integrated into a single analytical process. The methodology consisted of data acquisition and cleaning, exploratory data analysis, visualization, statistical analysis, and machine learning model development and evaluation.

2.1 Data Acquisition and Initial Inspection

The Titanic dataset was obtained from Kaggle in CSV format. The dataset contained **891 records and 12 columns**. Python and Pandas were used to load the dataset and inspect its structure. Functions such as `head()`, `shape`, `info()`,

`describe()`, `isnull().sum()`, and `duplicated().sum()` were used to understand the dataset and identify data-quality issues.

2.2 Data Cleaning and Preparation

The dataset was examined for missing values and duplicate records. The analysis identified missing values in the **Age**, **Cabin**, and **Embarked** columns. Missing Age values were replaced using the median, missing Embarked values were replaced using the mode, and the Cabin column was removed because it contained a large number of missing observations. The dataset was also checked for duplicate records, and no duplicate rows were found.

These preprocessing steps were performed to make the dataset more suitable for exploratory analysis and subsequent modelling.

2.3 Exploratory Data Analysis

After cleaning the dataset, exploratory data analysis was performed to understand the distribution and relationships among important variables. Numerical summaries and graphical techniques were used to identify patterns in passenger characteristics and survival outcomes.

The analysis included variables such as passenger class, gender, age, fare, and embarkation. Visualizations were created using **Matplotlib** and **Seaborn** to make the observed patterns easier to interpret.

2.4 Data Visualization and Storytelling

Different visualizations were used to communicate the findings clearly. Survival was examined across gender and passenger class, while age and fare distributions were also explored.

Each visualization was interpreted in terms of:

- Purpose of the visualization
- Observed pattern
- Supporting numerical evidence
- Possible analytical insight

This approach helped connect the graphical results with the underlying data rather than presenting charts without interpretation.

2.5 Statistical Analysis and Hypothesis Testing

A statistical hypothesis was formulated to investigate whether passenger gender was associated with survival.

The hypotheses were:

H₀: There is no statistically significant association between passenger gender and survival.

H₁: There is a statistically significant association between passenger gender and survival.

A significance level of $\alpha = 0.05$ was selected. Since `sex` and `Survived` are categorical variables, the **Chi-Square Test of Independence** was selected.

A contingency table was created using the observed survival frequencies for male and female passengers. The `chi2_contingency()` function from SciPy was then used to calculate the Chi-Square statistic, p-value, degrees of freedom, and expected frequencies.

The test produced a Chi-Square statistic of approximately **260.717**, with **1 degree of freedom**, and a p-value of approximately 1.197×10^{-58} . The p-value was compared with the predefined significance level to determine the statistical decision.

2.6 Machine Learning Model Development

The statistical analysis was followed by a predictive modelling stage. A **Logistic Regression** model was developed to predict the `Survived` variable.

The selected input features were:

- **Pclass**
- **Sex**
- **Age**
- **Fare**

The target variable was **Survived**. Missing rows in the selected modelling variables were removed, and the categorical `Sex` variable was converted into numerical form before model training. After preprocessing, **714 records** remained for modelling.

The dataset was divided into **80% training data and 20% testing data**. The model was trained on the training set and evaluated on unseen test data.

2.7 Model Evaluation

The Logistic Regression model was evaluated using multiple performance measures rather than accuracy alone. A **confusion matrix** was used to examine correct and incorrect predictions, while a **ROC curve and AUC** were used to evaluate the model's classification ability.

The model achieved approximately **75.52% accuracy** and an **AUC of 0.81** on the evaluation data. These results were treated as the baseline performance of the selected Logistic Regression approach.

2.8 Integrated Analysis

Finally, the findings from data cleaning, visualization, statistical testing, and machine learning were considered

together. This integration helped distinguish between descriptive patterns, statistically tested associations, and predictive model performance.

The combined findings were then used to identify key insights, formulate strategic recommendations, discuss potential business or research impacts, and identify limitations and areas for future improvement.

3. Detailed Analysis and Key Findings

The analysis performed during the internship was divided into four major stages: data preparation and exploratory analysis, visualization and storytelling, statistical analysis, and machine learning. The findings from each stage were integrated to develop a complete understanding of the Titanic dataset.

3.1 Week 1 — Data Acquisition, Cleaning and Exploratory Analysis

The first stage focused on acquiring, cleaning, and understanding the Titanic dataset. The dataset contained **891 passenger records and 12 columns**.

The initial data-quality analysis identified missing values in several variables. The **Age** column contained **177 missing values**, while **Embarked** contained **2 missing values**. The **Cabin** column contained **687 missing values**, making it unsuitable for direct analysis. Age values were replaced using the median, missing Embarked values were filled using the mode, and the Cabin column was removed. No duplicate records were identified.

The exploratory analysis also examined passenger demographics, survival distribution, and relationships among numerical variables. The correlation analysis provided additional information about relationships such as Pclass and Fare and the relationship between SibSp and Parch.

Key finding:

The first stage demonstrated that proper data cleaning and preprocessing are essential before performing statistical analysis or machine learning.

3.2 Week 2 — Data Visualization and Storytelling

The second stage focused on transforming the cleaned data into meaningful visual information. Multiple visualizations were created using **Matplotlib and Seaborn**.

The analysis examined survival patterns according to passenger gender and passenger class. Additional visualizations were used to understand fare and age distributions and other passenger characteristics.

The visual analysis showed noticeable differences in survival outcomes across passenger groups. In particular, gender and passenger class provided useful patterns that could be investigated further through statistical analysis.

Key finding:

Visualization made it easier to identify patterns that might not be immediately visible from raw numerical data. It also provided a foundation for selecting meaningful questions for further statistical investigation.

3.3 Week 3 — Statistical Analysis and Hypothesis Testing

The third stage investigated whether the observed difference in survival between male and female passengers was statistically significant.

A **Chi-Square Test of Independence** was applied to the `Sex` and `Survived` variables. The observed contingency table contained:

Gender Did Not Survive Survived

Female	81	233
Male	468	109

The test produced the following results:

Measure	Result
Chi-Square Statistic	260.717
Degrees of Freedom	1
p-value	1.197×10^{-58}
Significance Level	0.05

Since the p-value was substantially smaller than 0.05, the null hypothesis was rejected. The analysis therefore provided statistical evidence of an association between passenger gender and survival in the dataset.

Descriptive analysis showed that **233 out of 314 female passengers survived**, giving an observed survival rate of approximately **74.2%**. Among male passengers, **109 out of 577 survived**, giving an observed survival rate of approximately **18.9%**. The difference between the observed

survival proportions was approximately **55.3 percentage points**.

Key finding:

The statistical analysis supported the visual observation that survival outcomes differed substantially between the gender categories. However, the statistical association should not be interpreted as proof of causation.

3.5 Integrated Key Findings

Combining the findings from all four stages produced the following major insights:

1. **Data quality is an important part of the data science workflow.** Missing values and unsuitable variables were identified and handled before analysis.
2. **Exploratory visualization revealed meaningful survival patterns** across passenger characteristics.
3. **Statistical testing provided evidence** that gender and survival were associated in the analyzed dataset.
4. **Female passengers had an observed survival rate of approximately 74.2%, compared with 18.9% for male passengers.**
5. **Logistic Regression achieved 75.52% accuracy and an AUC of 0.81** using Pclass, Sex, Age, and Fare.
6. The results demonstrate the value of combining **descriptive analysis, statistical testing, visualization, and predictive modelling** rather than relying on only one analytical technique.

4. Dataset Information

The dataset used for this project is the **Titanic Dataset**, which was downloaded from the **Kaggle** website. The original dataset contains **891 passenger records and 12 columns**.

The dataset contains information about passengers, including their survival status, passenger class, gender, age, ticket fare, family information, cabin information, and port of embarkation.

Important Variables

Column	Description
PassengerId	Unique passenger identification number
Survived	Survival status: 0 = Did not survive, 1 = Survived
Pclass	Passenger class
Name	Passenger name
Sex	Passenger gender
Age	Passenger age
SibSp	Number of siblings or spouses aboard
Parch	Number of parents or children aboard
Ticket	Ticket information
Fare	Passenger fare
Cabin	Cabin information
Embarked	Port of embarkation

For the machine learning model, the main features selected were **Pclass, Sex, Age, and Fare**, while **Survived** was used as the target variable.

5. Week 1 – Data Cleaning and Exploratory Data Analysis

5.1 Data Acquisition

The Titanic dataset was downloaded from the **Kaggle** website and loaded into Python using the Pandas library.

```
df = pd.read_csv("Titanic-Dataset.csv")
```

After loading the dataset, basic functions such as `head()`, `shape`, `info()`, and `describe()` were used to understand the dataset structure and its statistical characteristics.

5.2 Data Cleaning

The dataset was checked for missing values and other data quality issues.

Missing values were mainly observed in the **Age**, **Cabin**, and **Embarked** columns.

The following approaches were used:

- Missing values in **Age** were handled using the median age.
- Missing values in **Embarked** were handled using the mode.
- The **Cabin** column was removed because it contained a large number of missing values.
- The dataset was checked for duplicate records.
- Important columns and their data types were reviewed.

The cleaning process helped prepare the dataset for further analysis.

5.3 Exploratory Data Analysis

After cleaning the dataset, exploratory analysis was performed using Pandas and visualization libraries.

The analysis focused on:

- Passenger survival
- Gender
- Passenger class
- Age
- Fare
- Embarkation port

Key Findings

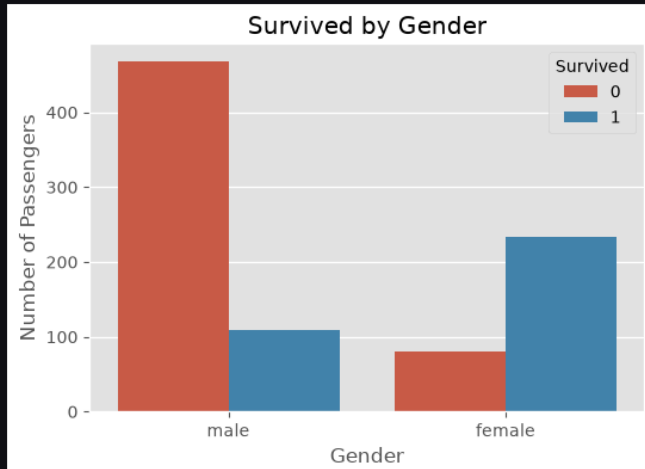
The exploratory analysis showed that:

1. Female passengers had a considerably higher survival count than male passengers.
2. First Class passengers had relatively better survival outcomes.
3. Third Class passengers had a much larger number of non-survivors.
4. Many passengers were in the younger and middle-age groups.
5. Most passengers paid relatively lower fares, while a small number paid significantly higher fares.

5.4 Visualizations

Survival by Gender

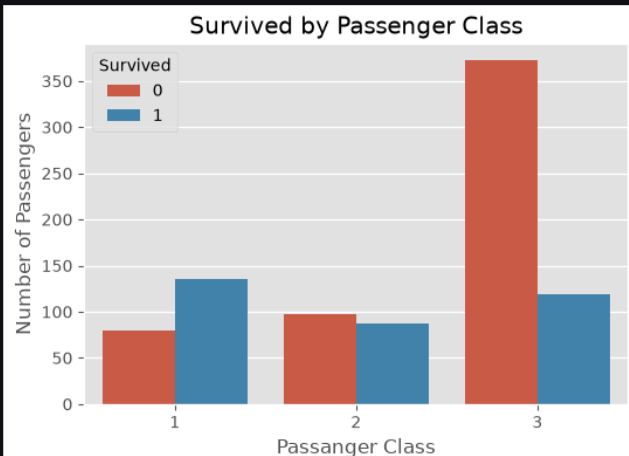
```
plt.figure(figsize=(6,4))
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survived by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()
```



Insight: The visualization shows a clear difference in survival outcomes between male and female passengers.

Survival by Passenger Class

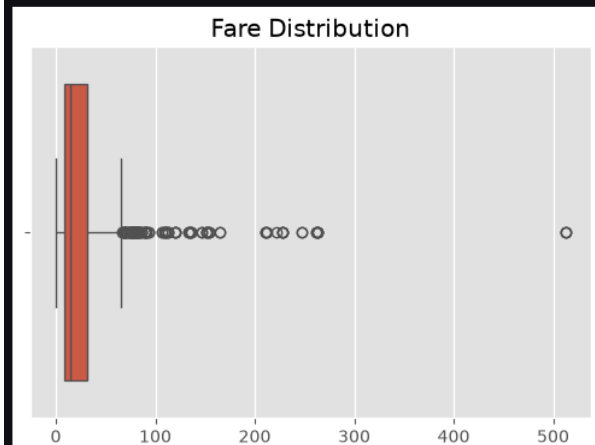
```
plt.figure(figsize=(6, 4))
sns.countplot(x = "Pclass", hue="Survived", data=df)
plt.title("Survived by Passenger Class")
plt.xlabel("Passanger Class")
plt.ylabel("Number of Passengers")
plt.show()
```



Insight: First Class passengers had relatively better survival outcomes, while Third Class had a larger number of non-survivors.

Fare Distribution

```
plt.figure(figsize=(6,4))
sns.boxplot(x = df["Fare"])
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.show()
```



Insight: Most passengers paid relatively lower fares, while a small number of passengers paid considerably higher fares.

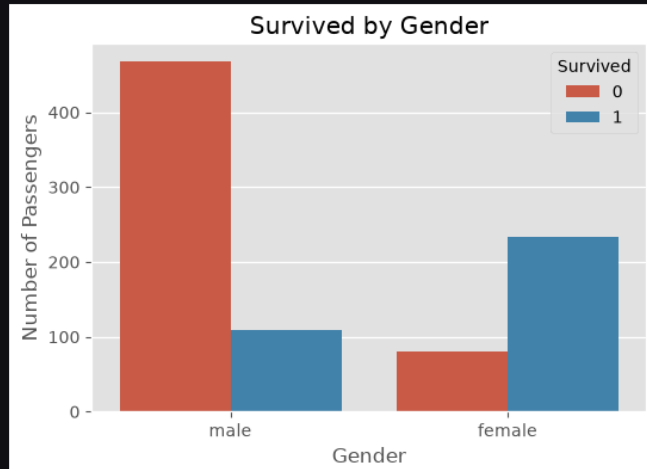
6. Week 2 – Advanced Data Visualization and Storytelling

During Week 2, the focus was on creating visualizations that could communicate important patterns and trends in the Titanic dataset clearly. Matplotlib and Seaborn were used to create different types of charts.

The purpose was not only to create graphs but also to understand what each visualization tells us about the data.

6.1 Survival by Gender

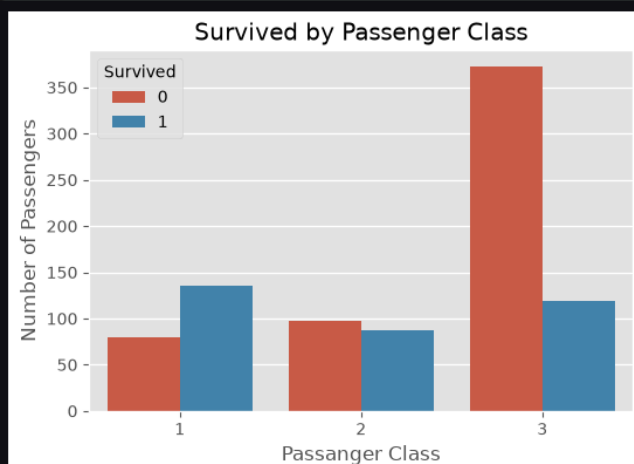
```
plt.figure(figsize=(6,4))
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survived by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()
```



Insight: The graph shows a clear difference in survival outcomes between male and female passengers. Female passengers had a considerably higher survival count than male passengers.

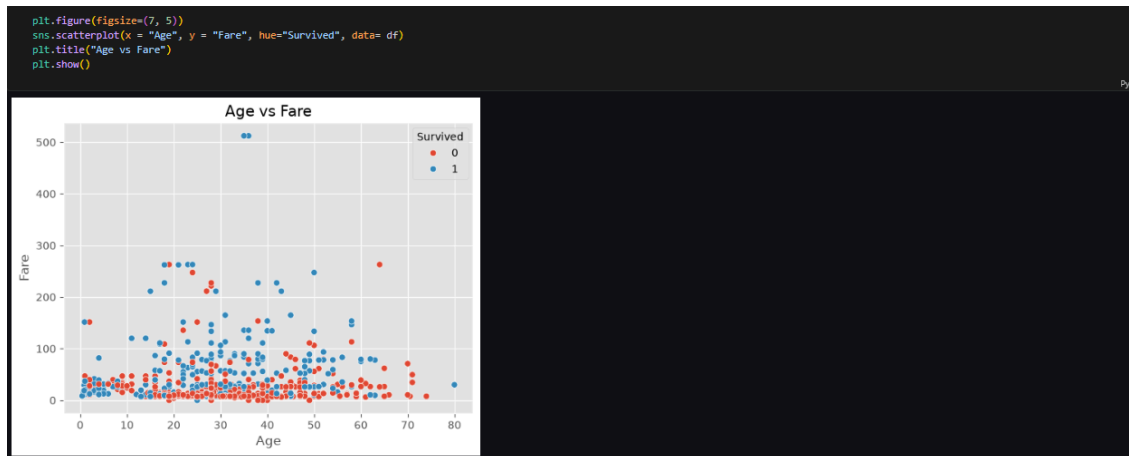
6.2 Survival by Passenger Class

```
plt.figure(figsize=(6, 4))
sns.countplot(x = "Pclass", hue="Survived", data=df)
plt.title("Survived by Passenger Class")
plt.xlabel("Passanger Class")
plt.ylabel("Number of Passengers")
plt.show()
```



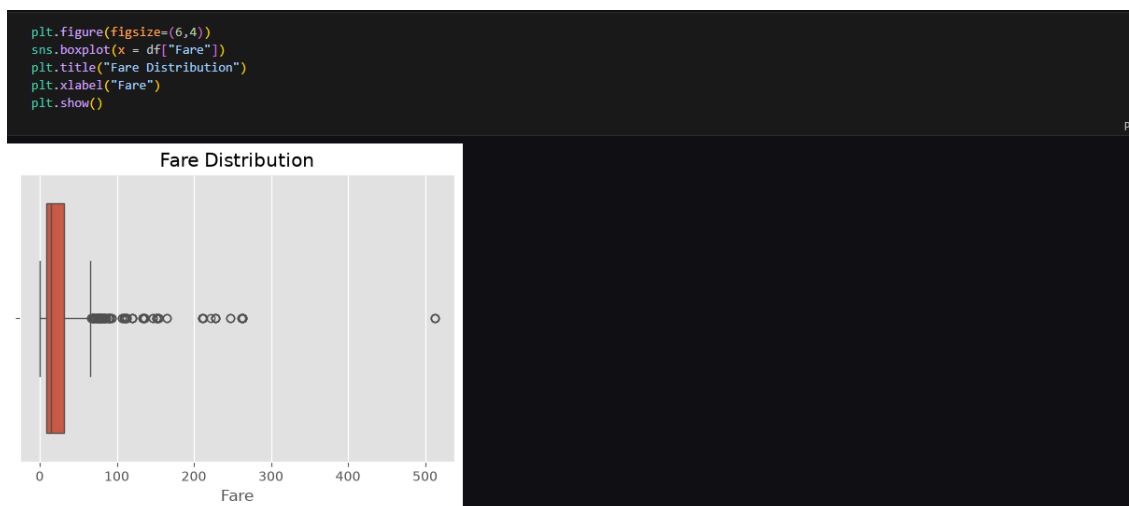
Insight: Survival outcomes differed across passenger classes. First Class passengers had relatively better survival outcomes, while Third Class passengers had a much larger number of non-survivors.

6.3 Age vs Fare



Insight: The scatter plot shows the distribution of passenger age and fare. Passengers of different ages paid a wide range of fares, and there was no strong direct relationship visible between age and fare.

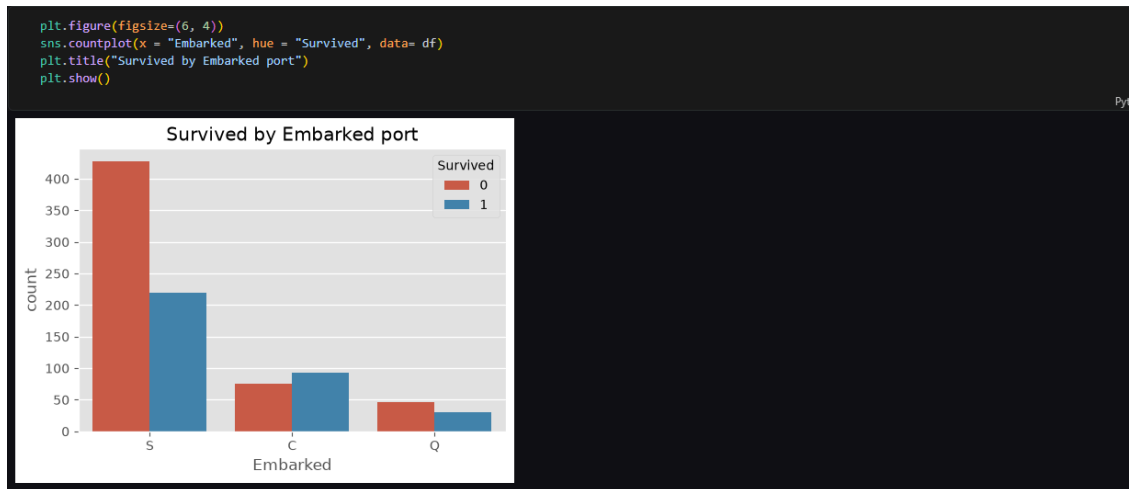
6.4 Fare Distribution



Insight: The fare distribution shows that most passengers paid relatively lower fares, while a smaller number of passengers paid

much higher fares. These higher values can be seen as high-value observations in the distribution.

6.5 Survival by Embarked Port



Insight: The visualization compares survival outcomes according to the passenger's embarkation port. Southampton had the largest number of passengers, while Cherbourg and Queenstown had fewer passengers.

6.6 Overall Visualization Findings

The visualizations made it easier to identify important patterns in the Titanic dataset. The differences in survival by gender and passenger class were particularly noticeable.

These visual findings provided a foundation for the statistical analysis performed in Week 3 and the machine learning analysis performed in Week 4.

```
# Week 3
## Statistical Analysis and Hypothesis Testing in python
**Dataset:** Titanic Dataset (Kaggle)
**Hypothesis:** Female passenger had a higher survival rate than male
passanger.
```

Hypothesis

Null Hypothesis(H0) There is no significant relationship between passanger gender and survival

Alternative Hypothesis (H1) There is a significant relationship between passanger gender and survival

```
table = pd.crosstab(df["Sex"], df["Survived"])

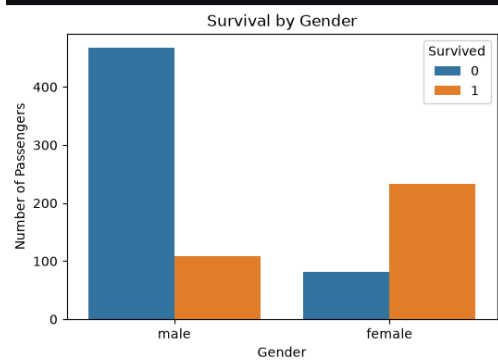
table
```

Survived	0	1
Sex		
female	81	233
male	408	109

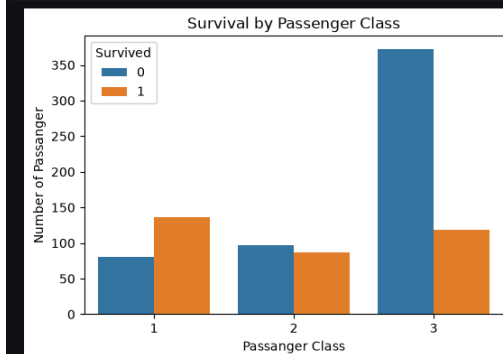
```
chi2, p, dof, expected = chi2_contingency(table)
print("Chi_Square Value:", chi2)
print("P-value:", p)
print("Degrees of Freedom", dof)
print("Expected Frequencies:")
print(expected)
```

Chi_Square Value: 269.71702916732184
P-value: 1.1973578627755645e-58
Degrees of Freedom: 1
Expected Frequencies:
[[193.47474747 128.52525253]
 [355.52525253 221.47474747]]

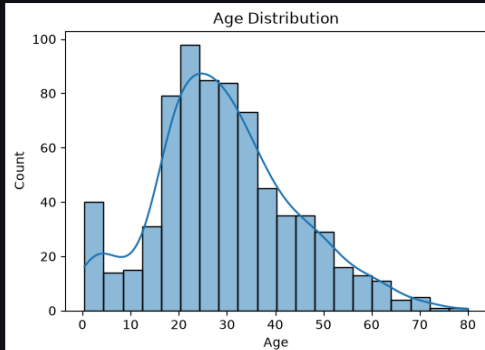
```
plt.figure(figsize=(6,4))
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()
```



```
plt.figure(figsize=(6,4))
sns.countplot(x="Pclass", hue="Survived", data=df)
plt.title("Survival by Passenger Class")
plt.xlabel("Passanger Class")
plt.ylabel("Number of Passanger")
plt.show()
```



```
plt.figure(figsize=(6,4))
sns.histplot(df["Age"], bins = 20, kde= True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



8. Week 4 – Machine Learning Model Development and Evaluation

During Week 4, a machine learning model was developed using the Titanic dataset to predict whether a passenger survived or did not survive. The complete process included data preparation, feature selection, train-test splitting, model training, prediction, and evaluation.

8.1 Feature Selection

The following features were selected for the machine learning model:

- **Pclass** – Passenger class
- **Sex** – Passenger gender
- **Age** – Passenger age
- **Fare** – Passenger fare

The target variable was:

- **Survived** – 0 = Did not survive, 1 = Survived

The `sex` column was converted into numerical values:

- Male = 0
- Female = 1

8.2 Train-Test Split

The dataset was divided into training and testing data using an 80:20 ratio.

Approximately 80% of the data was used to train the model, while 20% was used to evaluate its performance on unseen data.

8.3 Model Selection

Logistic Regression was selected because the target variable has two possible outcomes: survived or did not survive. Therefore, it is suitable for this binary classification problem.

```
model = LogisticRegression()  
model.fit(X_train, y_train)
```

8.4 Prediction

After training, the model was used to predict the survival status of passengers in the test dataset.

```
y_pred = model.predict(X_test)
```

8.5 Model Accuracy

The Logistic Regression model achieved an accuracy of approximately:

75.52%

This means that the model correctly predicted approximately 75.52% of the survival outcomes in the test dataset.

8.6 Confusion Matrix

A confusion matrix was created to compare the actual survival status with the predicted survival status.

[INSERT YOUR CONFUSION MATRIX SCREENSHOT HERE]

Interpretation

The confusion matrix shows the number of correct and incorrect predictions made by the model. The diagonal values represent correct predictions, while the other values represent incorrect predictions.

8.7 ROC Curve

A ROC curve was created to evaluate the model's ability to distinguish between passengers who survived and those who did not.

[INSERT YOUR ROC CURVE SCREENSHOT HERE]

AUC Result

AUC = 0.81

An AUC value of 0.81 indicates that the Logistic Regression model has a reasonable ability to distinguish between the two survival classes.

8.8 Overall Machine Learning Finding

The Logistic Regression model provided a reasonable baseline for predicting passenger survival. However, some predictions were incorrect, indicating that additional features and different machine learning algorithms could be explored to improve performance.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, roc_curve, auc

df = pd.read_csv("Titanic-Dataset.csv")
df.head()
```

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
df = df[['Survived', 'Pclass', 'Sex', 'Age', 'Fare']]
df = df.dropna()
df.head()
```

Survived	Pclass	Sex	Age	Fare	
0	0	3	male	22.0	7.2500
1	1	1	female	38.0	71.2833
2	1	3	female	26.0	7.9250
3	1	1	female	35.0	53.1000
4	0	3	male	35.0	8.0500

```
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})
df.head()
```

Python

	Survived	Pclass	Sex	Age	Fare
0	0	3	0	22.0	7.2500
1	1	1	1	38.0	71.2833
2	1	3	1	26.0	7.9250
3	1	1	1	35.0	53.1000
4	0	3	0	35.0	8.0500

```
df = pd.read_csv("Titanic-Dataset.csv")

df = df[['Survived', 'Pclass', 'Sex', 'Age', 'Fare']]

df = df.dropna()

df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})

df.head()
```

Python

	Survived	Pclass	Sex	Age	Fare
0	0	3	0	22.0	7.2500
1	1	1	1	38.0	71.2833
2	1	3	1	26.0	7.9250
3	1	1	1	35.0	53.1000
4	0	3	0	35.0	8.0500

```
X = df[['Pclass', 'Sex', 'Age', 'Fare']]
y = df['Survived']
print("X shape:", X.shape)
print("y shape:", y.shape)
```

Python

X shape: (714, 4)
y shape: (714,)

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)
```

Python

Training data: (571, 4)
Testing data: (143, 4)

```
#LogisticRegression Model Train
model = LogisticRegression()
model.fit(X_train, y_train)
print("Model training completed.")
```

Python

Model training completed.

```
#prediction
y_pred = model.predict(X_test)
print("Predictions:")
print(y_pred[:10])
```

Python

Predictions:
[0 1 1 0 0 1 1 1]

```
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")
```

Python

Accuracy: 0.7552447552447552
Accuracy Percentage: 75.52447552447552 %

```
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

Python

Classification Report:

	precision	recall	f1-score	support
0	0.81	0.78	0.80	87
1	0.68	0.71	0.70	56
accuracy			0.76	143
macro avg	0.74	0.75	0.75	143
weighted avg	0.76	0.76	0.76	143

```
cm = confusion_matrix(y_test, y_pred)

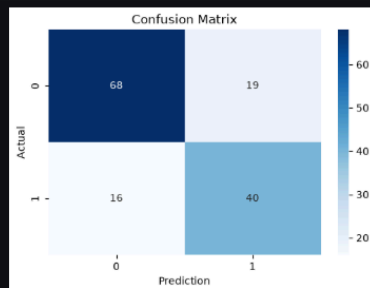
print("Confusion Matrix:")
print(cm)
```

Python

Confusion Matrix:
[[68 19]
 [16 40]]

```
plt.figure(figsize=(5, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.xlabel("Prediction")
plt.ylabel("Actual")
plt.show()
```

Python



```

y_prob = model.predict_proba(X_test)[:, 1]

fpr, tpr, thresholds = roc_curve(y_test, y_prob)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6, 4))

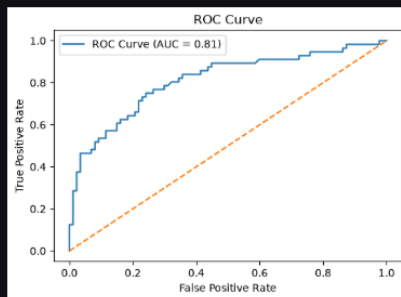
plt.plot(fpr, tpr, label=f"ROC Curve (AUC = {roc_auc:.2f})")
plt.plot([0, 1], [0, 1], linestyle="--")

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()

plt.show()

print("AUC:", roc_auc)

```



AUC: 0.8109685911338049

```

print("Model Evaluation Summary")
print(".....")
print("Accuracy:", accuracy)
print("Accuracy (%)", accuracy * 100)
print("AUC:", roc_auc)

```

Python

```

Model Evaluation Summary
.....
Accuracy: 0.7552447552447552
Accuracy (%): 75.52447552447552
AUC: 0.8109685911338049

```

Model Evaluation and Discussion

Strengths

The Logistic Regression model achieved an accuracy of approximately 75.52% and an AUC of 0.81 on the test dataset. The ROC curve shows that the model has a reasonable ability to distinguish between passengers who survived and those who did not. The model is also simple, fast, and easy to interpret.

Limitations

The model does not predict every passenger correctly. Some passengers were classified incorrectly, as shown in the confusion matrix. The Titanic dataset also contains missing information and limited features, so the model cannot capture every factor that may have influenced survival.

Possible Improvements

The model could be improved by using additional useful features such as Family Size, SibSp, Parch, and Embarked. Different algorithms such as Decision Tree, Random Forest, or Gradient Boosting could also be tested and compared. Feature scaling and hyperparameter tuning may further improve model performance.

9. Overall Key Findings

The complete analysis of the Titanic dataset produced several important findings across the different stages of the project.

9.1 Data Analysis Findings

The exploratory analysis showed clear differences in survival outcomes among passenger groups. Female passengers had a considerably higher survival count than male passengers. Passenger

class also showed noticeable differences, with First Class passengers having relatively better survival outcomes than Third Class passengers.

9.2 Statistical Findings

The Chi-Square Test of Independence produced a Chi-Square value of approximately **260.72** and a p-value of approximately 1.20×10^{-58} .

Since the p-value was much smaller than 0.05, the Null Hypothesis was rejected. This indicates a statistically significant association between passenger gender and survival in the Titanic dataset.

9.3 Machine Learning Findings

The Logistic Regression model was developed using Pclass, Sex, Age, and Fare as features. The model achieved an accuracy of approximately **75.52%** and an **AUC of 0.81**.

The model demonstrated a reasonable ability to distinguish between passengers who survived and those who did not.

10. Strategic Recommendations

Based on the findings obtained from data cleaning, visualization, statistical analysis, and machine learning, the following recommendations can be proposed. These recommendations are intended as data-driven decision-support suggestions rather than direct conclusions for modern transportation systems.

10.1 Use Multiple Passenger Characteristics

The analysis showed differences in survival outcomes across gender and passenger class. The machine learning model also used **Pclass, Sex, Age, and Fare** as predictive features. Therefore, future analysis should consider multiple relevant variables together rather than depending on a single characteristic.

10.2 Combine Statistical and Predictive Analysis

Statistical testing and machine learning provide different types of information. The Chi-Square test identified a statistically significant association between gender and survival, while Logistic Regression provided a predictive approach with **75.52% accuracy and 0.81 AUC**. Combining these approaches can provide both statistical evidence and predictive information.

10.3 Improve the Machine Learning Model

The current model provides a baseline but uses only four features. Future versions can include additional variables such as **SibSp, Parch, Embarked, and family-related features**. Different algorithms such as Decision Tree, Random Forest, and Gradient Boosting can also be evaluated and compared.

10.4 Use Cross-Validation and Hyperparameter Tuning

The current model was evaluated using an 80:20 train-test split. Future analysis should use cross-validation and hyperparameter tuning to obtain a more reliable estimate of model performance and identify suitable model parameters.

10.5 Use Visualizations for Stakeholder Communication

The project showed that visualizations made differences in survival outcomes easier to understand. Charts such as survival by gender, survival by passenger class, fare distribution, and survival by embarkation port can be used to communicate analytical findings clearly to non-technical stakeholders.

10.6 Improve Data Quality Before Analysis

Data quality should be checked before performing statistical or machine learning analysis. Missing values, duplicate records, inappropriate data types, and irrelevant columns should be identified and handled using a documented methodology.

10.7 Use Larger and More Relevant Datasets

The Titanic dataset is useful for demonstrating data science techniques, but it is historical. For real-world transportation or safety applications, organizations should use larger, recent, and domain-specific datasets containing relevant operational and passenger information.

10.8 Treat Machine Learning as Decision Support

The Logistic Regression model achieved **75.52% accuracy and an AUC of 0.81**, but it still produced incorrect predictions. Therefore, a predictive model should be treated as a decision-support tool rather than the only basis for important decisions.

10.9 Continuously Evaluate Analytical Systems

Data analysis and predictive models should be evaluated regularly when new data becomes available. Model performance, data quality, and relevance of selected features should be reviewed so that analytical systems can be improved over time.

11. Business Impact and Research Outcomes

The project demonstrates how a complete data science workflow can support evidence-based analysis and decision-making. The analysis combined data cleaning, exploratory analysis, visualization, statistical hypothesis testing, and machine learning to study survival patterns in the Titanic dataset.

From a decision-support perspective, the analysis identified clear differences in survival outcomes across passenger groups. The Chi-Square Test of Independence produced a Chi-Square value of approximately **260.72** with a p-value of approximately **1.20×10^{-58}** , indicating a statistically

significant association between passenger gender and survival in this dataset. The visualization analysis also showed noticeable differences in survival outcomes across gender and passenger class.

The machine learning stage demonstrated how predictive analytics can be used to estimate outcomes from available passenger information. A Logistic Regression model using **Pclass, Sex, Age, and Fare** achieved **75.52% accuracy** and an **AUC of 0.81**. This provides a baseline for understanding how selected variables can be used in a classification problem.

The project also has research value because it demonstrates the importance of combining different analytical methods rather than depending on a single technique. Visualization helps identify patterns, statistical testing evaluates whether an observed relationship is statistically significant, and machine learning demonstrates predictive modeling.

However, the Titanic dataset is historical and should not be directly used for modern transportation or safety decisions. Its primary business and research value in this project is as a learning and methodology example. For real-world applications, organizations would require larger, recent, domain-specific, and higher-quality datasets.

Overall, the project demonstrates how data science can transform raw data into measurable findings, statistical evidence, predictive results, and structured recommendations that can support informed decision-making.

12. Limitations

Although the project provides a complete data science workflow, several limitations should be considered when interpreting the results.

1. Historical Dataset:

The Titanic dataset represents a historical event and may not represent the conditions, passenger behavior, safety systems, or operational procedures of modern transportation systems. Therefore, the findings should not be directly generalized to present-day transportation.

2. Missing Data:

The original dataset contained missing values, particularly in the Age, Cabin, and Embarked columns. Age values were handled using median imputation, Embarked values using mode imputation, and the Cabin column was removed because of substantial missing information. These preprocessing decisions may influence the analysis.

3. Limited Machine Learning Features:

The Logistic Regression model used only **Pclass, Sex, Age, and Fare** as input features. Other available variables such as SibSp, Parch, Embarked, and family-related information were not included in the model.

4. Single Machine Learning Algorithm:

Only Logistic Regression was evaluated. Other algorithms such as Decision Tree, Random Forest, Gradient Boosting, or Support Vector Machine were not compared in this project.

5. Train-Test Split:

The model used an **80:20 train-test split**. A single split may not provide a complete picture of model performance across different samples.

6. Model Performance:

The Logistic Regression model achieved **75.52% accuracy**

and an AUC of 0.81. Although these results provide a useful baseline, the model still produced incorrect predictions. Therefore, it should not be considered a perfect predictive system.

7. Statistical Scope:

The statistical analysis primarily focused on the relationship between gender and survival using the Chi-Square Test of Independence. Other possible relationships between variables were not tested statistically in the same depth.

8. Generalization:

The conclusions are specific to the available Titanic dataset. Further validation using larger, more recent, and domain-specific datasets would be required before applying similar methods to real-world decision-making.

13. Future Improvements

The current project provides a basic but complete data science workflow. The following improvements can make the analysis more detailed, reliable, and useful for future studies.

1. Feature Engineering:

New features can be created from the existing variables. For example, family-related information can be combined to create a family-size feature. This may provide additional information for predictive analysis.

2. Include Additional Features:

The machine learning model can be extended by including variables such as **SibSp, Parch, Embarked, and other relevant passenger information** instead of using only Pclass, Sex, Age, and Fare.

3. Compare Multiple Machine Learning Algorithms:

Future work can compare Logistic Regression with algorithms such as

Decision Tree, Random Forest, Gradient Boosting, and Support Vector Machine to determine how different approaches perform on the same dataset.

4. Cross-Validation:

Instead of depending on a single train-test split, cross-validation can be used to evaluate model performance across multiple subsets of the data.

5. Hyperparameter Tuning:

Hyperparameters of the selected machine learning algorithms can be optimized using suitable techniques. This may help improve model performance.

6. Additional Statistical Tests:

Future analysis can investigate other relationships in the dataset using appropriate statistical tests. This would provide a broader statistical understanding of the factors associated with survival.

7. Larger and More Recent Datasets:

Since the Titanic dataset is historical, future research should use larger and more recent datasets when studying modern transportation or safety-related problems.

8. Interactive Dashboard:

The analysis could be presented through an interactive dashboard using tools such as **Power BI, Tableau, or Streamlit**. This would make the findings easier for stakeholders to explore.

9. Better Model Evaluation:

Future work can include additional evaluation measures such as precision, recall, F1-score, ROC-AUC, and cross-validation results to provide a more comprehensive assessment of model performance.

10. Real-World Validation:

Before applying similar analytical methods to real-world situations, the model and findings should be validated using domain-specific data and appropriate operational conditions.

14. Final Conclusion

This five-week project provided a complete end-to-end experience of a data science workflow using the Titanic dataset obtained from Kaggle. The project integrated data acquisition, data cleaning, exploratory data analysis, visualization, statistical hypothesis testing, and machine learning into a single analytical study.

During the initial stage, the dataset was examined and prepared for analysis. Missing values in important columns were handled using appropriate preprocessing techniques, while the Cabin column was removed because of its large number of missing values. Exploratory analysis and visualizations were then used to identify patterns in passenger survival across characteristics such as gender, passenger class, age, fare, and embarkation port.

The statistical analysis provided quantitative evidence for one of the observed patterns. The Chi-Square Test of Independence produced a Chi-Square value of approximately **260.72** and a p-value of approximately 1.20×10^{-58} . Since the p-value was below the significance level of 0.05, the Null Hypothesis was rejected, indicating a statistically significant association between gender and survival in the Titanic dataset.

The machine learning stage further demonstrated the use of predictive analytics. A Logistic Regression model was developed using **Pclass, Sex, Age, and Fare** as features. The model achieved an accuracy of **75.52%** and an **AUC of 0.81**, providing a baseline for the classification problem.

The overall project demonstrates that different data science techniques provide complementary insights. Exploratory analysis helps identify patterns, visualization makes those patterns easier to communicate, statistical testing provides evidence about relationships, and machine learning provides a predictive perspective.

At the same time, the findings must be interpreted within the limitations of the dataset and methodology. The Titanic dataset is historical, only selected features were used for prediction, and only one machine learning algorithm was evaluated. Therefore, the results should not be directly generalized to modern transportation or safety systems.

Overall, the project successfully demonstrates how raw data can be transformed into structured findings, statistical evidence, predictive results, and data-driven recommendations. The experience also provides a strong foundation for future work involving larger datasets, advanced machine learning algorithms, feature engineering, cross-validation, and interactive data visualization.

15. References

1. Kaggle – Titanic Dataset
2. Pandas Documentation
3. Matplotlib Documentation
4. Seaborn Documentation
5. SciPy Documentation
6. Scikit-learn Documentation